

Quantum Mechanics III

Lecture 1

Quantum States and Preparation Procedures

- Operational setting
- Mathematical representation
- Dynamics and measurement
- Consistency conditions
- Example

■ Operational setting

Preparation and measurement procedures are distinguished from their matrix representations.

$$\rho \geq 0, \text{Tr}\rho = 1$$

■ Mathematical representation

Positivity, trace, and tensor-product conventions are fixed before the main calculation.

$$p(k|P, M) = \text{Tr}(\rho_P E_k)$$

■ Dynamics and measurement

The map from a state to observed probabilities is written explicitly for the experiment under discussion.

$$\gamma = \text{Tr}(\rho^2)$$

■ Consistency conditions

Complete positivity, normalization, Hermiticity, and appropriate classical limits are checked as needed.

■ Example

The example is worked for a qubit or two-mode system before extending the notation to higher dimension.

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Reading: Park notes, ch. 1; QDC-26 intro; Nielsen–Chuang §2.4